Building Quality Control (QC) Inspection Report

[UNRESOLVED ISSUE: PROJECT SCOPE UNCERTAIN]

Project Name: Skyline Heights Apartment Complex - Phase 3
Project Location: 742 Evergreen Terrace, Springfield, IL 62704
Inspection Date: February 10, 2026
Report ID: QC-2026-02-SH-003
Lead Inspector: James R. Sterling, Senior QC Engineer
Contractor: Apex Construction Group LLC

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1. Executive Summary

This Quality Control (QC) Inspection Report details the findings of the Phase 3 strucural and finishing audit for the Skyline Heights Apartment Complex. The inspection focused on the strucural integrity of Building B (Floors 4-6), the installation of the building envelope, and the initial MEP (Mechanical, Electrical, Plumbing) rough-

ins for the residential units. This report are very important. [INSPECTOR COMMENTS HERE]

1.1 Project Overview

The Skyline Heights project is a multi-residential development consisting of three 5-story towers. [NOTE: CONFLICTING INFORMATION] Phase 3 encompasses the structural completion and enclosure of Tower B. The objective of this QC check is to ensure that all work performed by Apex Construction Group and its subcontractors meets the rigorous standards set forth in the project specifications and local building codes.

1.2 Overall Status (Preliminary)

Category	Status	Notes
Structural Integrity	Pass	All concrete tests within tolerance.
Building Envelope	Pass with Observations	Minor flashing issues noted on North elevation. [ACTION: FOLLOW UP]
MEP Systems	In Progress	Electrical rough-ins 90% complete; Plumbing at 75%. [STATUS: PENDING]
Safety Compliance	Pass	Site safety protocols strictly followed.
Finishes	N/A	Interior finishes scheduled for next phase.

1.2 Major Findings

1. **Concrete Compressive Strength: [UNVERIFIED DATA]** Cylinder tests for the 5th-floor slab averaged 4,500 PSI, exceeding the design requirement of 4,000 PSI.
2. **Window Flashing: [CRITICAL DEFECT]** Inconsistent application of flashing tape on three units (402, 405, 408). Immediate rectification required to prevent moisture ingress.
3. **Electrical Conduit: [MINOR ISSUE]** Minor blockage found in one vertical riser; cleared during the inspection.

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2. Inspection Methodology

The inspection was conducted using a combination of visual assessments, physical measurements, and non-destructive testing (NDT). We did a comprehensiv job. The following tools and standards were utilized during the process:

2.1 Tools and Equipment

- **Laser Level (Bosch GLL3-330CG):** Used for verifying floor flatness and vertical alignment of columns and window frames.
- **Moisture Meter (FLIR MR176):** Utilized to check for moisture levels in the subfloor and wall cavities.
- **Ultrasonic Thickness Gauge: [TOOL CALIBRATION DUE]** Employed for checking the thickness of fireproofing material on steel beams.
- **Digital Calipers:** Used for precise measurement of reinforcement bar spacing and diameter.

2.2 Standards and References

The inspection was performed in accordance with the following regulatory frameworks:

- **IBC 2021:** International Building Code for structural and safety requirements.
- **ACI 318:** Building Code Requirements for Structural Concrete.
- **NEC 2023:** National Electrical Code for all electrical rough-ins.
- **ASTM E2112:** Standard Practice for Installation of Exterior Windows, Doors, and Skylights.

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3. Structural & Building Envelope Inspection

2.1 Concrete and Masonry

The strucural inspection covered the 5th and 6th-floor vertical elements and the 5th-floor horizontal slab. This is a 3-story building.

Item	Requirement	Observation	Result
Reinforcement Placement	Per Structural Drawing S-203	Rebar spacing and cover verified. [SEE ATTACHMENT A]	Pass
Concrete Slump Test	4” - 6”	Recorded at 5.25” for Batch #402. [BATCH NUMBER]	Pass
Formwork Alignment	Within ¼” tolerance	Laser level verified verticality. [VERIFIED BY: JR]	Pass
Anchor Bolt Placement	Per Mechanical Spec M-101	Secured and aligned for HVAC units. [CHECK NEXT WEEK]	Pass

2.2 Building Envelope (Exterior)

Focus was placed on moisture protection and thermal insulation continuity.

- **Vapor Barrier: [INCOMPLETE]** Continuous application verified on floors 4 and 5. No punctures or tears observed.
- **Window Installation: [NEEDS RE-INSPECTION]** All windows on the South and West elevations are plumb and level. Seals were tested for air tightness.
- **Roofing Membrane: [PARTIAL INSPECTION]** Initial application on the lower terrace (Floor 4) was inspected. Overlap of 4 inches maintained as per manufacturer specifications.

> ***Observation QC-01:*** *Flashing at the head of window units on the North elevation (Units 402-408) shows insufficent lap length. This is a big problem. Action Required: Re-apply flashing to ensure a minimum 6-inch vertical overlap.*

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3. MEP Systems (Mechanical, Electrical, Plumbing)

3.1 Electrical Systems

Rough-in inspection for residential units and common corridors. We are almost done.

System	Component	Status	Verification Method
Distribution	Main Feeders [CHECKED]	Verified	Visual inspection of cable tray.
Branch Circuits	Unit Wiring	90% Complete [EXPECTED 100%]	Insulation resistance test pending.
Safety	Grounding	Pass	Continuity test performed on ground bus. [PASSED]
Fire Alarm	Smoke Detectors	Pre-wired [TO BE TESTED]	Boxes positioned per NFPA 72.

3.2 Plumbing & HVAC

Review of pressure tests and ductwork sealing.

- **Domestic Water Lines: [PRESSURE TEST PENDING]** Pressure tested at 150 PSI for 2 hours. No drop in pressure recorded.
- **Sanitary Drainage: [SLOPE VERIFICATION]** Slope verified at ¼" per foot for all horizontal runs.
- **HVAC Ductwork: [SEALING CHECK]** Sealing of joints with mastic verified. Hanger spacing meets SMACNA standards.

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4. Material Quality & Safety Compliance

4.1 Material Verification

All materials delivered to the site were cross-referenced with the approved submittals. [MISSING DOCUMENTATION]

Material	Manufacturer	Submittal Ref	Batch/Lot
Rebar (Grade 60)	Nucor Steel [VENDOR ID: 123]	SUB-STR-012	B-2026-X1
PVC Pipe (Sch 40)	Charlotte Pipe [VENDOR ID: 456]	SUB-PLM-045	L-9982-A
Windows (Low-E)	Pella Corp [VENDOR ID: 789]	SUB-ENV-088	Order #7731

4.2 Safety Audit

A walk-through was conducted to ensure complience with OSHA and site-specific safety plans. [AUDIT PENDING]

- **Fall Protection: [CRITICAL SAFETY]** Guardrails are secure on all open floor edges. Personal Fall Arrest Systems (PFAS) being used by workers on the building exterior.
- **Housekeeping: [NEEDS IMPROVEMENT]** Debris removal is being handled daily. Walkways are clear of obstructions.
- **PPE Compliance: [GOOD JOB]** 100% compliance observed for hard hats, high-visibility vests, and safety glasses.

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5. Deficiency Log & Corrective Actions

The following items require attention before the next scheduled inspection on February 18, 2026. These are the deficencies.

ID	Description	Location	Priority	Assigned To
D-001	Window flashing lap length < 6” [URGENT]	Units 402-408	High	Envelope Sub
D-002	Exposed wiring in Room 512 [MEDIUM PRIORITY]	Floor 5 Hall	Medium	Electrical Sub
D-003	Minor concrete honeycombing [LOW PRIORITY]	Beam B-405	Low	Concrete Sub

5.1 Recommendations

1. **Toolbox Talk: [SCHEDULE ASAP]** Conduct a session for the envelope team on flashing overlap requirements.
2. **Moisture Testing: [RECOMMENDED]** Schedule infrared thermography for the North elevation once the exterior cladding is complete.

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6. Sign-off and Approval (Final)

The work inspected on February 11, 2026, is generally in accordance with the project plans and specifications, with the exception of the deficiencies noted in Section 5.

Inspector Signature:
James R. Sterling
James R. Sterling, Senior QC Engineer
Date: February 11, 2026 [CONFIRMED]

Contractor Acknowledgement:
Michael V. Rossi
Michael V. Rossi, Project Manager, Apex Construction
Date: February 11, 2026 [CONFIRMED]

End of Report